

## SAFETY DATA SHEET

Revision Date 01-April-2024

Revision Number 3

### 1. Identification

|                             |                                                                           |
|-----------------------------|---------------------------------------------------------------------------|
| <b>Product Name</b>         | <b>Sulfur in Isooctane standard solution, Specpure®, 50 µg/g (0.005%)</b> |
| <b>Cat No. :</b>            | <b>41318</b>                                                              |
| <b>Synonyms</b>             | No information available                                                  |
| <b>Recommended Use</b>      | Laboratory chemicals.                                                     |
| <b>Uses advised against</b> | Food, drug, pesticide or biocidal product use.                            |

#### Details of the supplier of the safety data sheet

##### Company

##### **Importer/Distributor**

Fisher Scientific  
112 Colonnade Road,  
Ottawa, ON K2E 7L6,  
Canada  
Tel: 1-800-234-7437

##### **Emergency Telephone Number**

For information **US** call: 001-800-227-6701 / **Europe** call: +32 14 57 52 11

Emergency Number **US**:001-201-796-7100 / **Europe**: +32 14 57 52 99

**CHEMTREC** Tel. No. **US**:001-800-424-9300 / **Europe**:001-703-527-3887

### 2. Hazard(s) identification

#### Classification

**WHMIS 2015 Classification** Classified as hazardous under the Hazardous Products Regulations (SOR/2015-17)

|                                                         |            |
|---------------------------------------------------------|------------|
| <b>Flammable liquids</b>                                | Category 2 |
| <b>Skin Corrosion/Irritation</b>                        | Category 2 |
| <b>Serious Eye Damage/Eye Irritation</b>                | Category 2 |
| <b>Specific target organ toxicity (single exposure)</b> | Category 3 |
| Target Organs - Central nervous system (CNS).           |            |
| <b>Aspiration Toxicity</b>                              | Category 1 |

#### Label Elements

##### **Signal Word**

Danger

##### **Hazard Statements**

Highly flammable liquid and vapor

May be fatal if swallowed and enters airways

Causes skin irritation  
Causes serious eye irritation  
May cause drowsiness and dizziness



### **Precautionary Statements**

#### **Prevention**

Keep away from heat, hot surfaces, sparks, open flames and other ignition sources. No smoking  
Keep container tightly closed  
Ground/bond container and receiving equipment  
Avoid breathing dust/fume/gas/mist/vapors/spray  
Wash face, hands and any exposed skin thoroughly after handling  
Use only outdoors or in a well-ventilated area  
Wear protective gloves/protective clothing/eye protection/face protection  
Use non-sparking tools  
Take action to prevent static discharges

#### **Response**

IF SWALLOWED: Immediately call a POISON CENTER/doctor  
IF ON SKIN (or hair): Take off immediately all contaminated clothing. Rinse skin with water/ shower  
IF INHALED: Remove person to fresh air and keep comfortable for breathing  
IF IN EYES: Rinse cautiously with water for several minutes. Remove contact lenses, if present and easy to do. Continue rinsing  
Call a POISON CENTER/ doctor if you feel unwell  
Do NOT induce vomiting  
In case of fire: Use dry sand, dry chemical or alcohol-resistant foam to extinguish  
Take off contaminated clothing and wash it before reuse

#### **Storage**

Store locked up  
Store in a well-ventilated place. Keep container tightly closed

#### **Disposal**

Dispose of contents/container to an approved waste disposal plant

#### **Other Hazards**

Very toxic to aquatic life with long lasting effects

## **3. Composition/Information on Ingredients**

| <b>Component</b>       | <b>CAS-No</b> | <b>Weight %</b> |
|------------------------|---------------|-----------------|
| 2,2,4-Trimethylpentane | 540-84-1      | 99.95           |
| Butane, 1,1-thiobis-   | 544-40-1      | 0.05            |

## **4. First-aid measures**

#### **General Advice**

If symptoms persist, call a physician.

#### **Eye Contact**

Rinse immediately with plenty of water, also under the eyelids, for at least 15 minutes. Get medical attention.

#### **Skin Contact**

Wash off immediately with plenty of water for at least 15 minutes. If skin irritation persists, call a physician.

|                                        |                                                                                                                                                                                                     |
|----------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Inhalation</b>                      | Remove to fresh air. If not breathing, give artificial respiration. Get medical attention if symptoms occur. Risk of serious damage to the lungs (by aspiration).                                   |
| <b>Ingestion</b>                       | Clean mouth with water and drink afterwards plenty of water. Do NOT induce vomiting. Call a physician or poison control center immediately. If vomiting occurs naturally, have victim lean forward. |
| <b>Most important symptoms/effects</b> | Difficulty in breathing. Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting                                                        |
| <b>Notes to Physician</b>              | Treat symptomatically                                                                                                                                                                               |

## 5. Fire-fighting measures

|                                         |                                                   |
|-----------------------------------------|---------------------------------------------------|
| <b>Suitable Extinguishing Media</b>     | Water mist may be used to cool closed containers. |
| <b>Unsuitable Extinguishing Media</b>   | No information available                          |
| <b>Flash Point</b>                      | -7 °C / 19.4 °F                                   |
| <b>Method -</b>                         | No information available                          |
| <b>Autoignition Temperature</b>         | 415 °C / 779 °F                                   |
| <b>Explosion Limits</b>                 |                                                   |
| <b>Upper</b>                            | No data available                                 |
| <b>Lower</b>                            | No data available                                 |
| <b>Sensitivity to Mechanical Impact</b> | No information available                          |
| <b>Sensitivity to Static Discharge</b>  | No information available                          |

### Specific Hazards Arising from the Chemical

Flammable. Containers may explode when heated. Vapors may form explosive mixtures with air. Vapors may travel to source of ignition and flash back. Do not allow run-off from fire-fighting to enter drains or water courses.

### Hazardous Combustion Products

None known.

### Protective Equipment and Precautions for Firefighters

As in any fire, wear self-contained breathing apparatus pressure-demand, MSHA/NIOSH (approved or equivalent) and full protective gear.

### NFPA

|               |                     |                    |                         |
|---------------|---------------------|--------------------|-------------------------|
| <b>Health</b> | <b>Flammability</b> | <b>Instability</b> | <b>Physical hazards</b> |
| 3             | 3                   | 0                  | -                       |

## 6. Accidental release measures

|                                             |                                                                                                                                                                                                                                             |
|---------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Personal Precautions</b>                 | Ensure adequate ventilation. Use personal protective equipment as required. Remove all sources of ignition. Take precautionary measures against static discharges.                                                                          |
| <b>Environmental Precautions</b>            | Do not flush into surface water or sanitary sewer system. Do not allow material to contaminate ground water system. Prevent product from entering drains. Local authorities should be advised if significant spillages cannot be contained. |
| <b>Methods for Containment and Clean Up</b> | Soak up with inert absorbent material. Keep in suitable, closed containers for disposal. Remove all sources of ignition. Use spark-proof tools and explosion-proof equipment.                                                               |

## 7. Handling and storage

|                 |                                                                                                                                                                                   |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>Handling</b> | Wear personal protective equipment/face protection. Ensure adequate ventilation. Do not get in eyes, on skin, or on clothing. Avoid ingestion and inhalation. Keep away from open |
|-----------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|

flames, hot surfaces and sources of ignition. Use only non-sparking tools. To avoid ignition of vapors by static electricity discharge, all metal parts of the equipment must be grounded. Take precautionary measures against static discharges.

**Storage.**

Keep container tightly closed in a dry and well-ventilated place. Keep away from heat, sparks and flame.

## 8. Exposure controls / personal protection

### Exposure Guidelines

| Component              | Alberta                                     | British Columbia | Ontario TWAEV | Quebec | ACGIH TLV    | OSHA PEL | NIOSH |
|------------------------|---------------------------------------------|------------------|---------------|--------|--------------|----------|-------|
| 2,2,4-Trimethylpentane | TWA: 300 ppm<br>TWA: 1400 mg/m <sup>3</sup> | TWA: 300 ppm     |               |        | TWA: 300 ppm |          |       |

**Legend**

ACGIH - American Conference of Governmental Industrial Hygienists

### Engineering Measures

Ensure adequate ventilation, especially in confined areas. Use explosion-proof electrical/ventilating/lighting/equipment.  
Wherever possible, engineering control measures such as the isolation or enclosure of the process, the introduction of process or equipment changes to minimise release or contact, and the use of properly designed ventilation systems, should be adopted to control hazardous materials at source

### Personal protective equipment

**Eye Protection**

Wear appropriate protective eyeglasses or chemical safety goggles as described by OSHA's eye and face protection regulations in 29 CFR 1910.133 or European Standard EN166.

**Hand Protection**

Wear appropriate protective gloves and clothing to prevent skin exposure.

| Glove material | Breakthrough time | Glove thickness | Glove comments         |
|----------------|-------------------|-----------------|------------------------|
| Natural rubber | See manufacturers | -               | Splash protection only |
| Nitrile rubber | recommendations   |                 |                        |
| Neoprene       |                   |                 |                        |
| PVC            |                   |                 |                        |

Inspect gloves before use. observe the instructions regarding permeability and breakthrough time which are provided by the supplier of the gloves. (Refer to manufacturer/supplier for information) gloves are suitable for the task: Chemical compatability, Dexterity, Operational conditions, User susceptibility, e.g. sensitisation effects, also take into consideration the specific local conditions under which the product is used, such as the danger of cuts, abrasion. gloves with care avoiding skin contamination.

**Respiratory Protection**

When workers are facing concentrations above the exposure limit they must use appropriate certified respirators. Follow the OSHA respirator regulations found in 29 CFR 1910.134 or European Standard EN 149. Use a NIOSH/MSHA or European Standard EN 149 approved respirator if exposure limits are exceeded or if irritation or other symptoms are experienced. To protect the wearer, respiratory protective equipment must be the correct fit and be used and maintained properly

**Recommended Filter type:** Particulates filter conforming to EN 143

When RPE is used a face piece Fit Test should be conducted

### Environmental exposure controls

Prevent product from entering drains. Do not allow material to contaminate ground water system. Local authorities should be advised if significant spillages cannot be contained.

### Hygiene Measures

Handle in accordance with good industrial hygiene and safety practice. Keep away from food, drink and animal feeding stuffs. Do

not eat, drink or smoke when using this product. Remove and wash contaminated clothing and gloves, including the inside, before re-use. Wash hands before breaks and after work.

## 9. Physical and chemical properties

|                                        |                               |
|----------------------------------------|-------------------------------|
| Physical State                         | Liquid                        |
| Appearance                             | Colorless                     |
| Odor                                   | Petroleum distillates         |
| Odor Threshold                         | No information available      |
| pH                                     | No information available      |
| Melting Point/Range                    | -107 °C / -160.6 °F           |
| Boiling Point/Range                    | 98 - 99 °C / 208.4 - 210.2 °F |
| Flash Point                            | -7 °C / 19.4 °F               |
| Evaporation Rate                       | No information available      |
| Flammability (solid,gas)               | Not applicable                |
| Flammability or explosive limits       |                               |
| Upper                                  | No data available             |
| Lower                                  | No data available             |
| Vapor Pressure                         | 117 hPa @ 38 °C               |
| Vapor Density                          | No information available      |
| Specific Gravity                       | 0.692 g/cm3                   |
| Solubility                             | No information available      |
| Partition coefficient; n-octanol/water | No data available             |
| Autoignition Temperature               | 415 °C / 779 °F               |
| Decomposition Temperature              | No information available      |
| Viscosity                              | No information available      |

## 10. Stability and reactivity

|                                  |                                                                   |
|----------------------------------|-------------------------------------------------------------------|
| Reactive Hazard                  | None known, based on information available                        |
| Stability                        | Stable under normal conditions.                                   |
| Conditions to Avoid              | Keep away from open flames, hot surfaces and sources of ignition. |
| Incompatible Materials           | Strong oxidizing agents                                           |
| Hazardous Decomposition Products | None under normal use conditions                                  |
| Hazardous Polymerization         | Hazardous polymerization does not occur.                          |
| Hazardous Reactions              | None under normal processing.                                     |

## 11. Toxicological information

### Acute Toxicity

#### Product Information

**Oral LD50** Based on ATE data, the classification criteria are not met. ATE > 2000 mg/kg.

**Dermal LD50** Based on ATE data, the classification criteria are not met. ATE > 2000 mg/kg.

**Vapor LC50** Based on ATE data, the classification criteria are not met. ATE > 20 mg/l.

#### Component Information

| Component              | LD50 Oral                 | LD50 Dermal         | LC50 Inhalation               |
|------------------------|---------------------------|---------------------|-------------------------------|
| 2,2,4-Trimethylpentane | LD50 5000 mg/kg ( Rat )   | 2000 mg/kg (Rabbit) | LC50 = 33.52 mg/L ( Rat ) 4 h |
| Butane, 1,1-thiobis-   | LD50 = 2220 mg/kg ( Rat ) | Not listed          | Not listed                    |

**Toxicologically Synergistic** No information available

#### Products

**Delayed and immediate effects as well as chronic effects from short and long-term exposure**

**Irritation** No information available

**Sensitization** No information available

**Carcinogenicity** The table below indicates whether each agency has listed any ingredient as a carcinogen.

| Component              | CAS-No   | IARC       | NTP        | ACGIH      | OSHA       | Mexico     |
|------------------------|----------|------------|------------|------------|------------|------------|
| 2,2,4-Trimethylpentane | 540-84-1 | Not listed | Not listed | Not listed | Not listed | Not listed |
| Butane, 1,1-thiobis-   | 544-40-1 | Not listed | Not listed | Not listed | Not listed | Not listed |

**Mutagenic Effects** No information available

**Reproductive Effects** No information available.

**Developmental Effects** No information available.

**Teratogenicity** No information available.

**STOT - single exposure** Central nervous system (CNS)

**STOT - repeated exposure** None known

**Aspiration hazard** No information available

**Symptoms / effects, both acute and delayed** Inhalation of high vapor concentrations may cause symptoms like headache, dizziness, tiredness, nausea and vomiting

**Endocrine Disruptor Information** No information available

**Other Adverse Effects** The toxicological properties have not been fully investigated.

## 12. Ecological information

### Ecotoxicity

Very toxic to aquatic organisms, may cause long-term adverse effects in the aquatic environment. The product contains following substances which are hazardous for the environment.

| Component              | Freshwater Algae     | Freshwater Fish                                                 | Microtox   | Water Flea                             |
|------------------------|----------------------|-----------------------------------------------------------------|------------|----------------------------------------|
| 2,2,4-Trimethylpentane | EC50= 2.94 mg/l, 72h | LC50 = 0.11 mg/l, 96h,<br>(Rainbow trout)                       | Not listed | EC50= 0.4 mg/l, 48h<br>(Daphnia magna) |
| Butane, 1,1-thiobis-   | Not listed           | LC50: = 3.58 mg/L, 96h<br>flow-through (Pimephales<br>promelas) | Not listed | Not listed                             |

**Persistence and Degradability** Immiscible with water Persistence is unlikely based on information available.

**Bioaccumulation/ Accumulation** No information available.

**Mobility** Is not likely mobile in the environment due its low water solubility. Will likely be mobile in the environment due to its volatility.

## 13. Disposal considerations

**Waste Disposal Methods** Chemical waste generators must determine whether a discarded chemical is classified as a hazardous waste. Chemical waste generators must also consult local, regional, and national hazardous waste regulations to ensure complete and accurate classification.

## 14. Transport information

### DOT

UN-No

UN1262

|                      |         |
|----------------------|---------|
| Proper Shipping Name | OCTANES |
| Hazard Class         | 3       |
| Packing Group        | II      |
| <b>TDG</b>           |         |
| UN-No                | UN1262  |
| Proper Shipping Name | OCTANES |
| Hazard Class         | 3       |
| Packing Group        | II      |
| <b>IATA</b>          |         |
| UN-No                | UN1262  |
| Proper Shipping Name | OCTANES |
| Hazard Class         | 3       |
| Packing Group        | II      |
| <b>IMDG/IMO</b>      |         |
| UN-No                | UN1262  |
| Proper Shipping Name | OCTANES |
| Hazard Class         | 3       |
| Packing Group        | II      |

## 15. Regulatory information

### International Inventories

| Component              | CAS-No   | DSL | NDSL | TSCA | TSCA Inventory notification - Active-Inactive | EINECS    | ELINCS | NLP |
|------------------------|----------|-----|------|------|-----------------------------------------------|-----------|--------|-----|
| 2,2,4-Trimethylpentane | 540-84-1 | X   | -    | X    | ACTIVE                                        | 208-759-1 | -      | -   |
| Butane, 1,1-thiobis-   | 544-40-1 | X   | -    | X    | ACTIVE                                        | 208-870-5 | -      | -   |

| Component              | CAS-No   | IECSC | KECL     | ENCS | ISHL | TCSI | AICS | NZIoC | PICCS |
|------------------------|----------|-------|----------|------|------|------|------|-------|-------|
| 2,2,4-Trimethylpentane | 540-84-1 | X     | KE-34634 | X    | X    | X    | X    | X     | X     |
| Butane, 1,1-thiobis-   | 544-40-1 | X     | -        | X    | X    | X    | X    | X     | X     |

#### Legend:

X - Listed '-' - Not Listed

KECL - NIER number or KE number (<http://ncis.nier.go.kr/en/main.do>)

DSL/NDSL - Canadian Domestic Substances List/Non-Domestic Substances List

TSCA - United States Toxic Substances Control Act Section 8(b) Inventory

EINECS/ELINCS - European Inventory of Existing Commercial Chemical Substances/EU List of Notified Chemical Substances

IECSC - Chinese Inventory of Existing Chemical Substances

KECL - Korean Existing and Evaluated Chemical Substances

ENCS - Japanese Existing and New Chemical Substances

AICS - Australian Inventory of Chemical Substances

PICCS - Philippines Inventory of Chemicals and Chemical Substances

### Canada

SDS in compliance with provisions of information as set out in Canadian Standard - Part 4, Schedule 1 and 2 of the Hazardous Products Regulations (HPR) and meets the requirements of the HPR (Paragraph 13(1)(a) of the Hazardous Products Act (HPA)).

| Component              | Canada - National Pollutant Release Inventory (NPRI) | Canadian Environmental Protection Agency (CEPA) - List of Toxic Substances | Canada's Chemicals Management Plan (CEPA) |
|------------------------|------------------------------------------------------|----------------------------------------------------------------------------|-------------------------------------------|
| 2,2,4-Trimethylpentane | Part 5, Isomer Groups Part 4 Substance               |                                                                            |                                           |

### Other International Regulations

#### Authorisation/Restrictions according to EU REACH

| Component | REACH (1907/2006) - Annex XIV - Substances Subject to | REACH (1907/2006) - Annex XVII - Restrictions on Certain Dangerous | REACH Regulation (EC 1907/2006) article 59 - Candidate |
|-----------|-------------------------------------------------------|--------------------------------------------------------------------|--------------------------------------------------------|
|-----------|-------------------------------------------------------|--------------------------------------------------------------------|--------------------------------------------------------|

|                        | Authorization | Substances                                                         | List of Substances of Very High Concern (SVHC) |
|------------------------|---------------|--------------------------------------------------------------------|------------------------------------------------|
| 2,2,4-Trimethylpentane | -             | Use restricted. See item 75.<br>(see link for restriction details) | -                                              |

**REACH links**

<https://echa.europa.eu/substances-restricted-under-reach>

**Safety, health and environmental regulations/legislation specific for the substance or mixture**

| Component              | CAS-No   | OECD HPV       | Persistent Organic Pollutant | Ozone Depletion Potential | Restriction of Hazardous Substances (RoHS) |
|------------------------|----------|----------------|------------------------------|---------------------------|--------------------------------------------|
| 2,2,4-Trimethylpentane | 540-84-1 | Listed         | Not applicable               | Not applicable            | Not applicable                             |
| Butane, 1,1-thiobis-   | 544-40-1 | Not applicable | Not applicable               | Not applicable            | Not applicable                             |

| Component              | CAS-No   | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Major Accident Notification | Seveso III Directive (2012/18/EC) - Qualifying Quantities for Safety Report Requirements | Rotterdam Convention (PIC) | Basel Convention (Hazardous Waste) |
|------------------------|----------|-------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|----------------------------|------------------------------------|
| 2,2,4-Trimethylpentane | 540-84-1 | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |
| Butane, 1,1-thiobis-   | 544-40-1 | Not applicable                                                                            | Not applicable                                                                           | Not applicable             | Not applicable                     |

## 16. Other information

**Prepared By**

Product Safety Department  
Email: chem.techinfo@thermofisher.com  
[www.thermofisher.com](http://www.thermofisher.com)

**Revision Date**

01-April-2024

**Print Date**

01-April-2024

**Revision Summary**

New emergency telephone response service provider.

**Disclaimer**

The information provided in this Safety Data Sheet is correct to the best of our knowledge, information and belief at the date of its publication. The information given is designed only as a guidance for safe handling, use, processing, storage, transportation, disposal and release and is not to be considered a warranty or quality specification. The information relates only to the specific material designated and may not be valid for such material used in combination with any other materials or in any process, unless specified in the text

**End of SDS**